



Management of Energy Efficiency and Safe Operation for Multi-Actuated Ground Vehicles on DWPT Roads

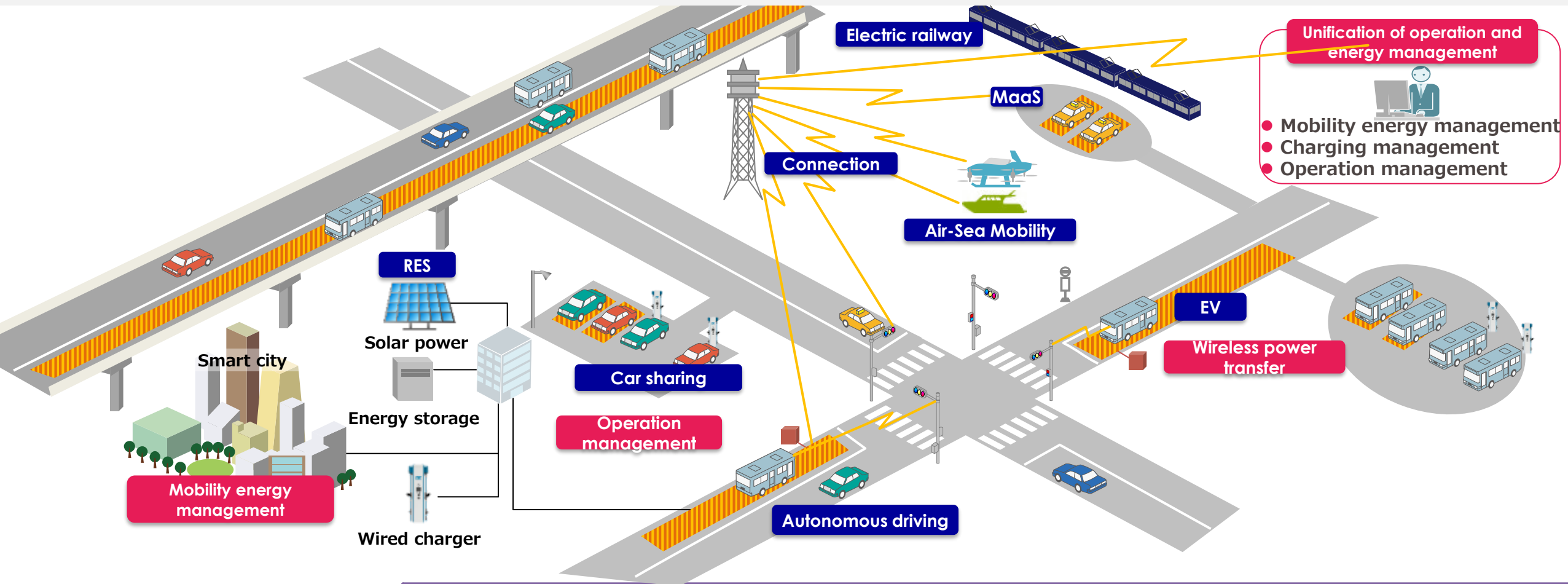
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¹The University of Tokyo, ²National University of Río Cuarto, ³Polytechnic University of Turin

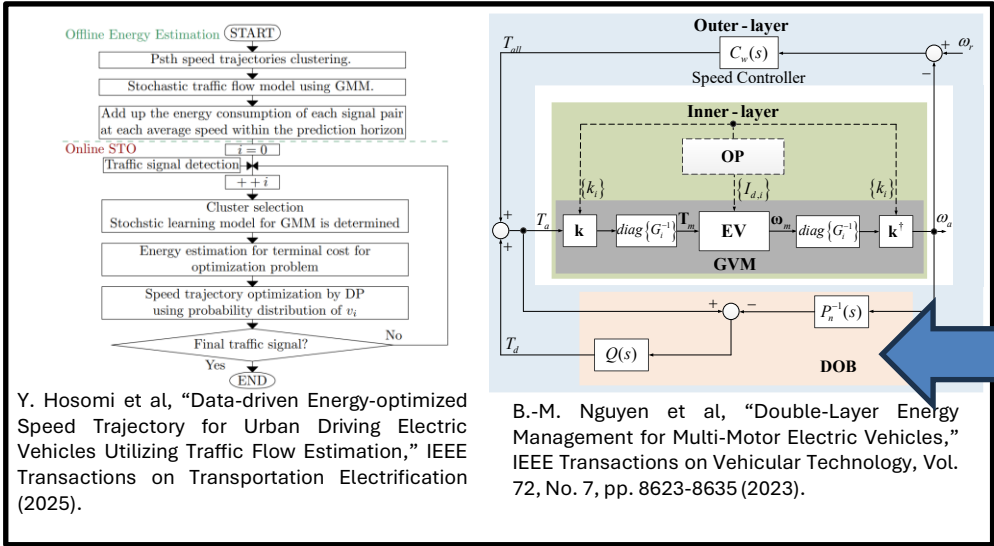
The future of e-Mobility

The transportation sector is undergoing a fundamental transformation driven by the urgent need to reduce carbon emissions and develop sustainable urban mobility systems.

- ❖ Emerging technologies such as *autonomous driving and motion control*, *optimal energy management and operation management*, and *dynamic wireless power transfer* are shaping the future of mobility.
- ❖ The development and integration of these systems demand not only technological innovation but also a *transdisciplinary approach* integrating expertise in electrical, automotive, control, mechanical, material engineering, etc.

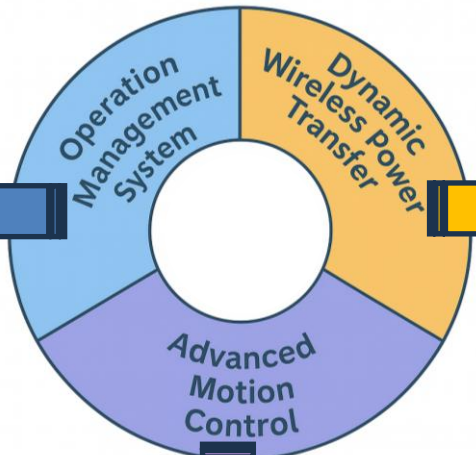


Our work toward the future of e-Mobility



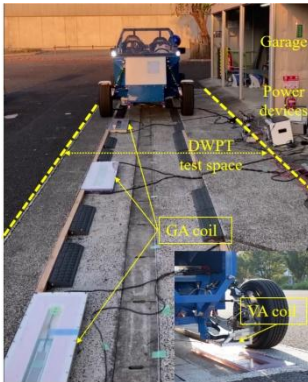
Speed trajectory generation

Torque allocation



T. Koishi et al, "Receiving Side Current-Based Lateral Misalignment Estimation and Automated Steering Control for Dynamic Wireless Power Transfer," IEEE Journal of Emerging and Selected Topics in Industrial Electronics, Vol. 6, No. 1, pp. 41-52 (2024).

Coil design



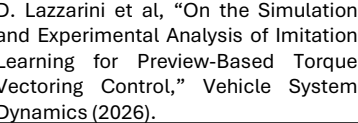
Position control



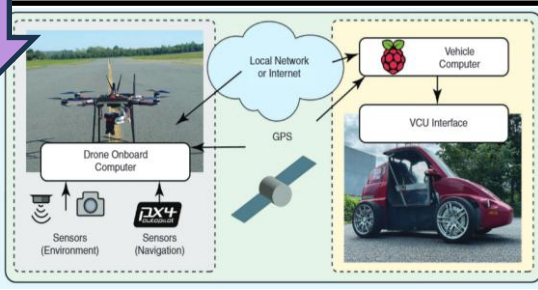
Slip ratio control

Friction force control

B.-M. Nguyen et al, "Slip Control for IWM Vehicles Based on Hierarchical LQR," Control Engineering Practice, Vol. 93, 104179 (2019).

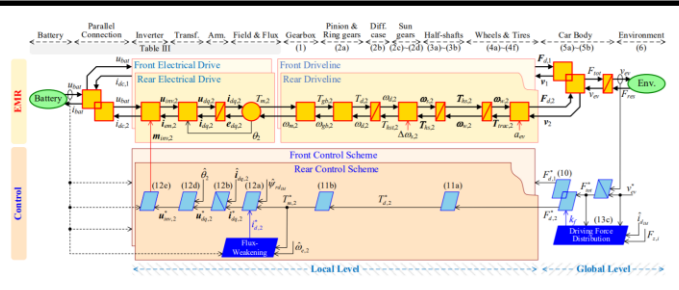


D. Lazzarini et al, "On the Simulation and Experimental Analysis of Imitation Learning for Preview-Based Torque Vectoring Control," Vehicle System Dynamics (2026).



Road condition estimation

V. Beliautso et al, "Drone-to-Vehicle Integration of Data: Design Concept and Application to Vehicle Automated System," IEEE Vehicular Technology Magazine (2025).

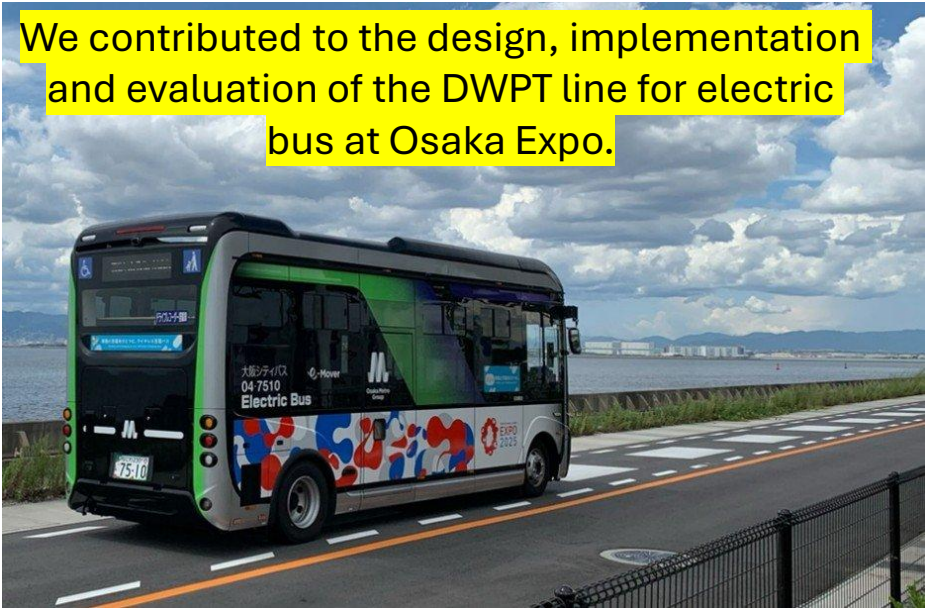


EMR Modeling of Vehicle

A. -T. Nguyen et al, "An Integrated Modeling Framework for Motion Control and Energy Management in Multi-Motor Electric Vehicles," IEEE Open Journal of Vehicular Technology (2025).

Our DWPT Development

We contributed to the design, implementation and evaluation of the DWPT line for electric bus at Osaka Expo.



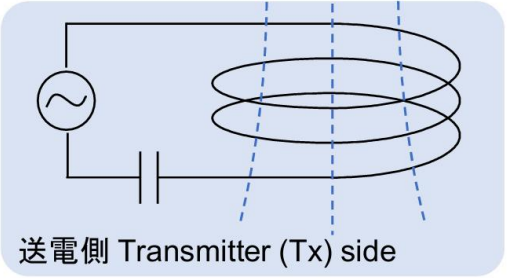
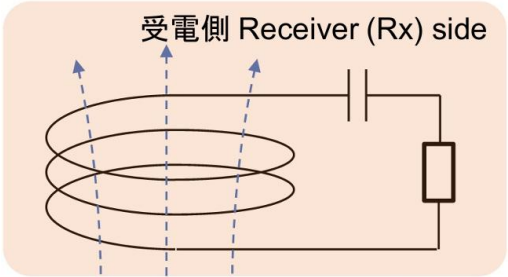
www.k.u-tokyo.ac.jp/information/category/press/11765.html



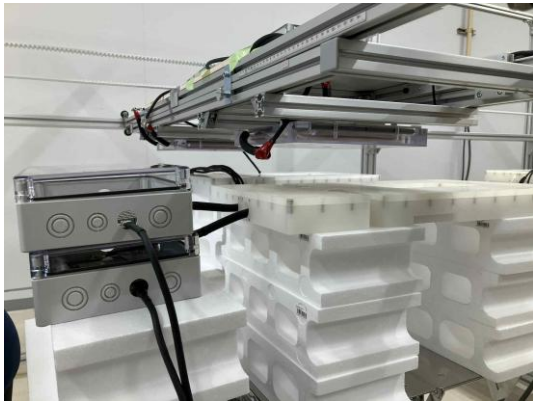
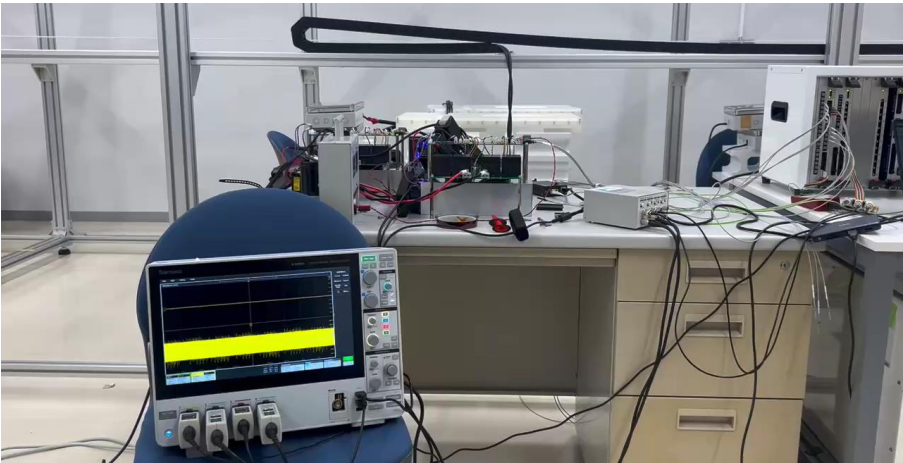
DWPT section at Kashiwanoha, Kashiwa City, Chiba Prefecture, Japan



Faraday: "Time varying magnetic flux induces electromotive force"



Ampere: "Electric current creates magnetic field around it"



Goal of IEEE VTS MVC 2027

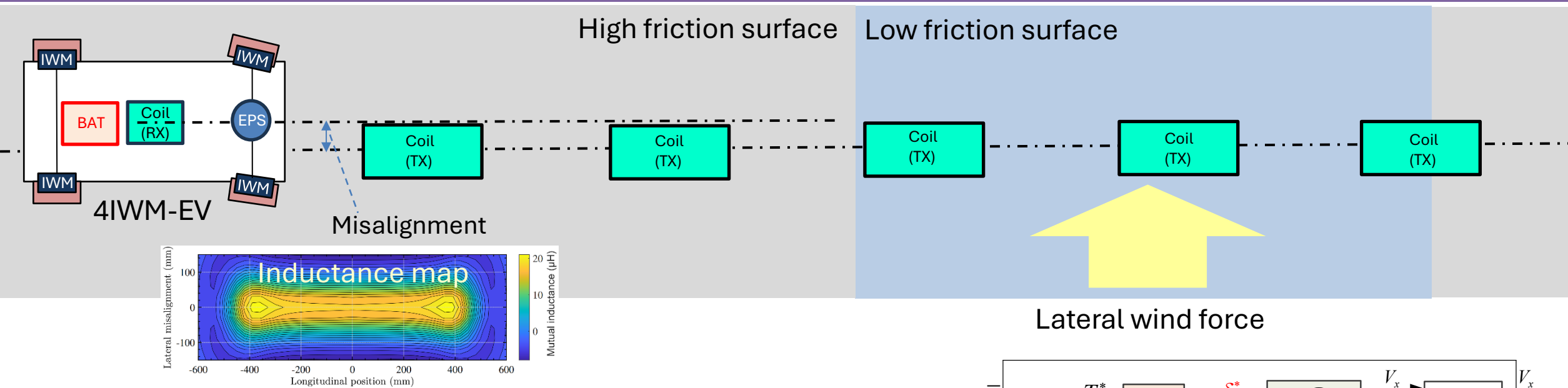
Management of Energy Efficiency and Safe Operation for Multi-Actuated Ground Vehicles on DWPT Roads

- ❖ Enhance the understanding of future e-Mobility concepts, enabling technologies, and key challenges.
- ❖ Demonstrate the need for a *transdisciplinary approach* to developing next-generation e-Mobility systems.
- ❖ Provide participants and the broader community with a multi-purpose simulator for vehicle dynamics education, evaluation of research ideas, and collaborative researches, etc.
- ❖ Attract participants from diverse academic and professional communities.

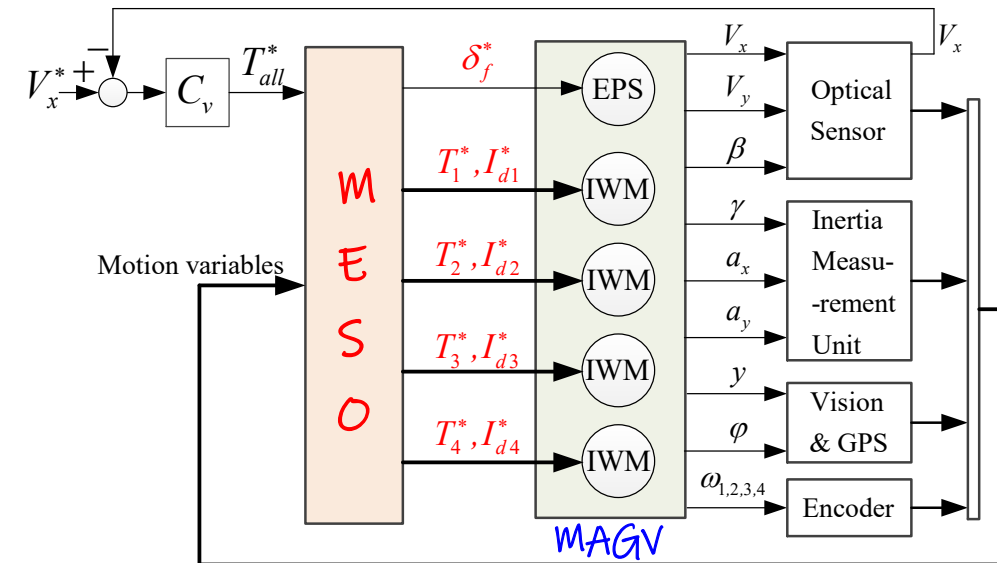
Remark: The simulation is based on a real experimental vehicle operated at the e-Mobility and Control Laboratory, The University of Tokyo. To ensure a feasible problem setting and maintain reasonable computational cost, certain dynamics and components of the real system are simplified or neglected. For example, the dynamics of the energy storage system and the power-electronics control, including coil detection and current control for the DWPT system, are not explicitly modeled. Nevertheless, the resulting model remains sufficiently detailed to provide a challenging simulation environment. The simulator has been developed using MATLAB R2022b, with a sampling time of 0.0002 s.



Problem Setting



- The electric vehicle is actuated by **4** in-wheel-motor and **1** front steering motor. The vehicle runs on a *straight* road. The speed pattern and speed controller are given.
- The vehicle may encounter a *low-friction* road surface and strong *lateral wind disturbances*.
- The DWPT is only available if the longitudinal and lateral misalignments are between certain thresholds. For instance, if the lateral misalignment exceeds 7cm, DWPT is shutdown.



Objectives

The participants are invited to design a strategy that generates {torque distribution ratios, motor d-axis currents, front steering angle} to attain multiple objectives:

(1) Drivetrain efficiency: Minimize the energy consumption:

$$\Phi_1 = \int_0^{t_{end}} P_{all} dt = \int_0^{t_{end}} \sum_{i=1}^4 (P_{m,i} + P_{cu,i} + P_{fe,i}) dt$$

(2) Safe traction: Minimize the wheel slip ratios:

$$\Phi_2 = \int_0^{t_{end}} \sum_{i=1}^4 \lambda_i^2 dt$$

(3) DWPT efficiency: Minimize DWPT efficiency penalty:

$$\Phi_3 = \int_0^{t_{end}} (1 - \eta_{DWPT})^2 dt$$

(4) Driving comfort: Minimize lateral acceleration:

$$\Phi_4 = \int_0^{t_{end}} a_y^2 dt$$

Scoring function: $\Phi = \sigma_1 \Phi_1 + \sigma_2 \Phi_2 + \sigma_3 \Phi_3 + \sigma_4 \Phi_4$

Constraints

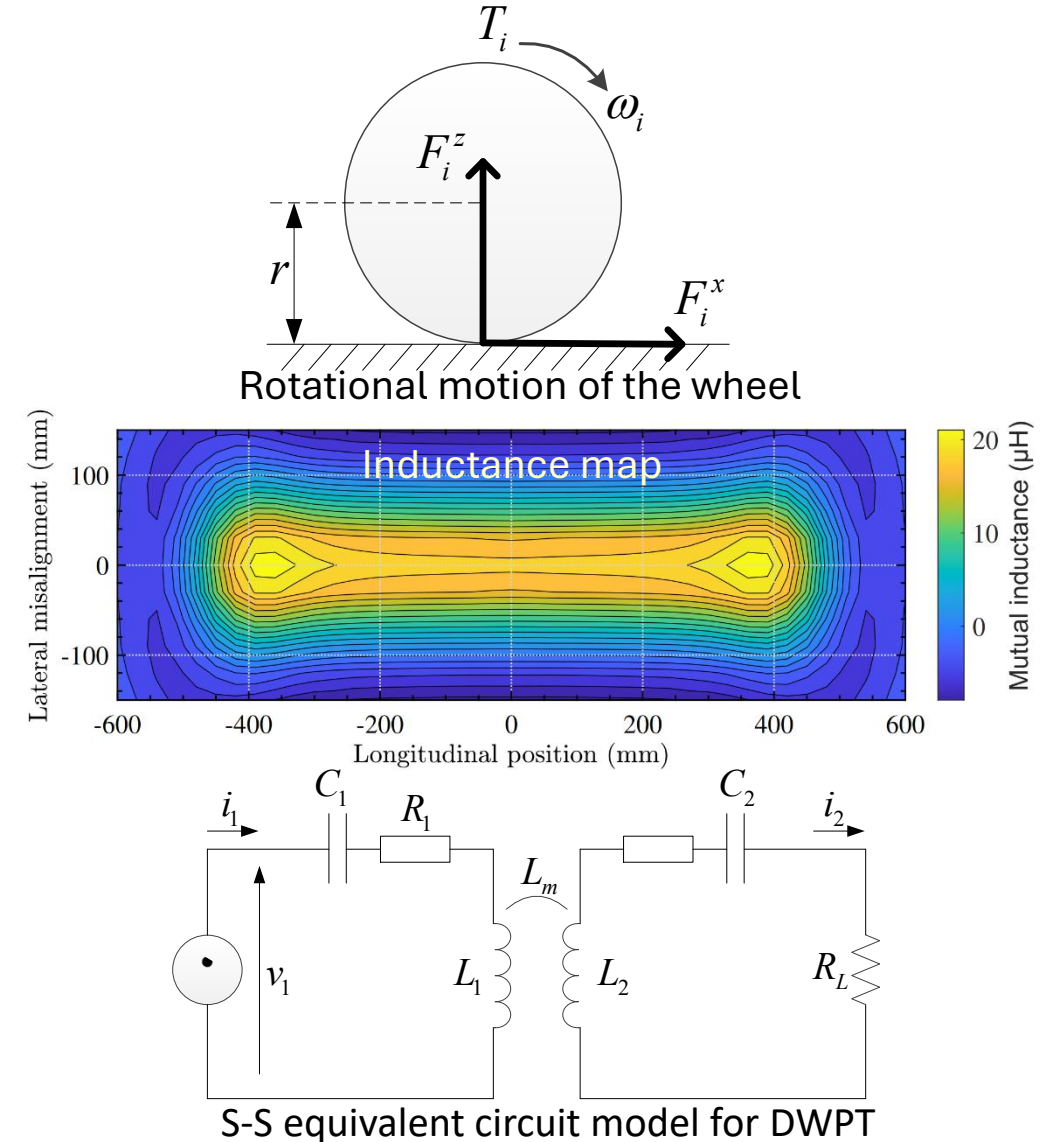
(1) **Total torque command:** $\sum_{i=1}^4 T_i^* = T_{all}^*$, where T_{all}^* is given by the velocity controller.

(2) **Motor torque limits:** $|T_i^*| \leq T_{max,i}, \forall i \in [1,4]$.

(3) **Friction limit circle:** $\sqrt{(F_i^x)^2 + (F_i^y)^2} \leq \mu F_i^z$,

(4) **Coil misalignment:** Wireless power transfer is enabled only when $|\Delta_x| \leq 380$ mm and $|\Delta_y| \leq 70$ mm, where Δ_x and Δ_y denote the longitudinal and lateral misalignments between TX and RX coils, respectively. This is to ensure sufficient mutual inductance and efficiency during power transfer operation:

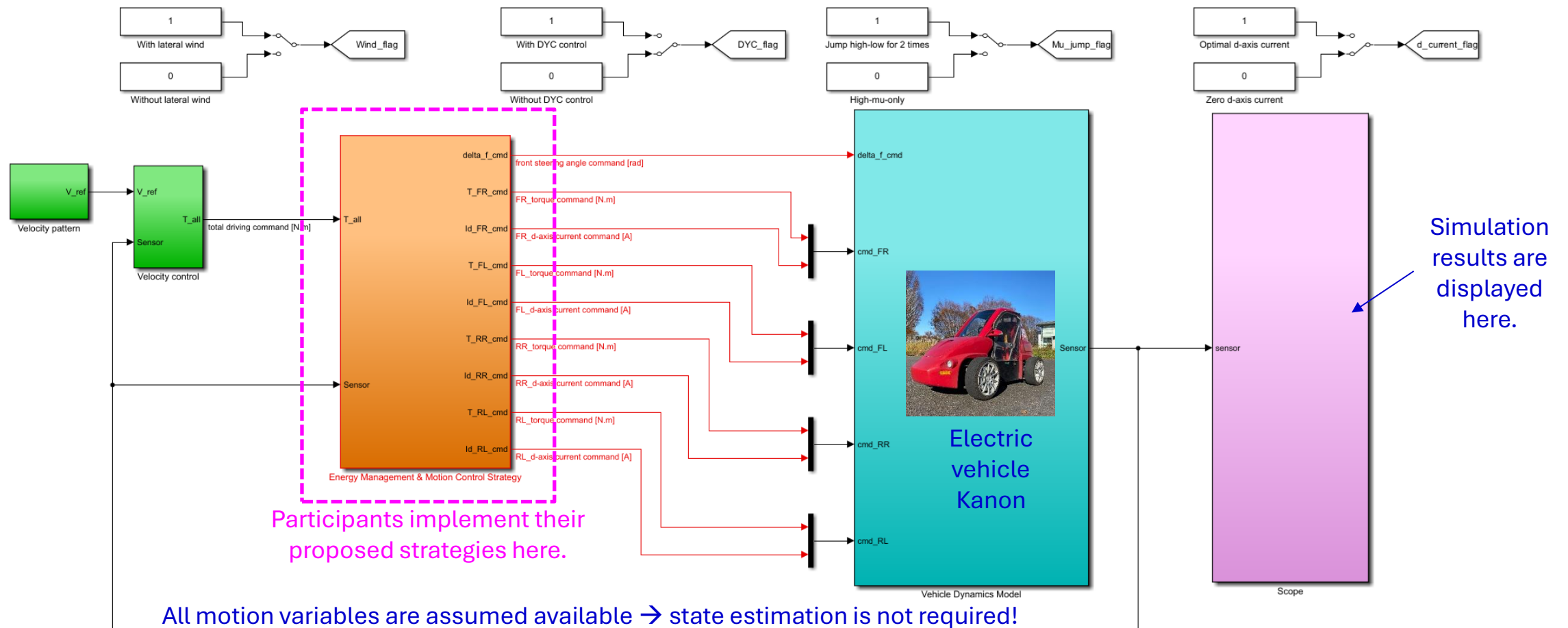
$$\eta_{DWPT} = \frac{(\omega_0 L_m)^2}{R_1 R_2 + R_1 R_L + (\omega_0 L_m)^2} \frac{R_L}{R_2 + R_L}$$



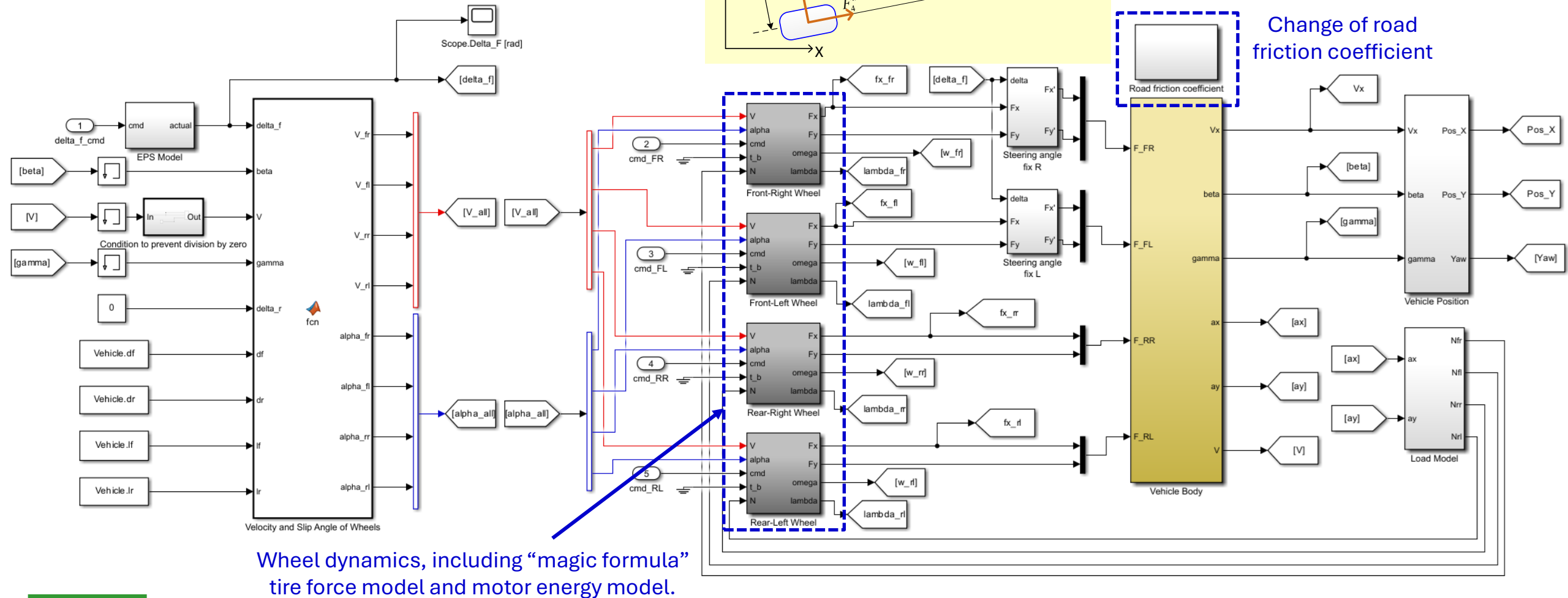
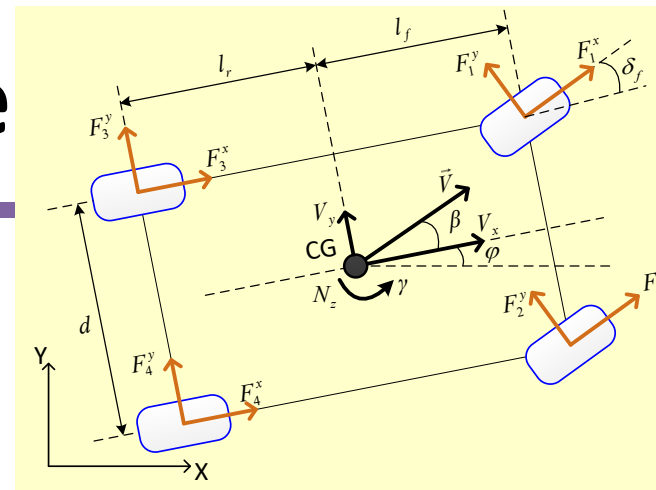
Simulation: Outline

Energy Management and Motion Control of an Electric Vehicle driven by Four In-Wheel-Motors with Active Front Steering

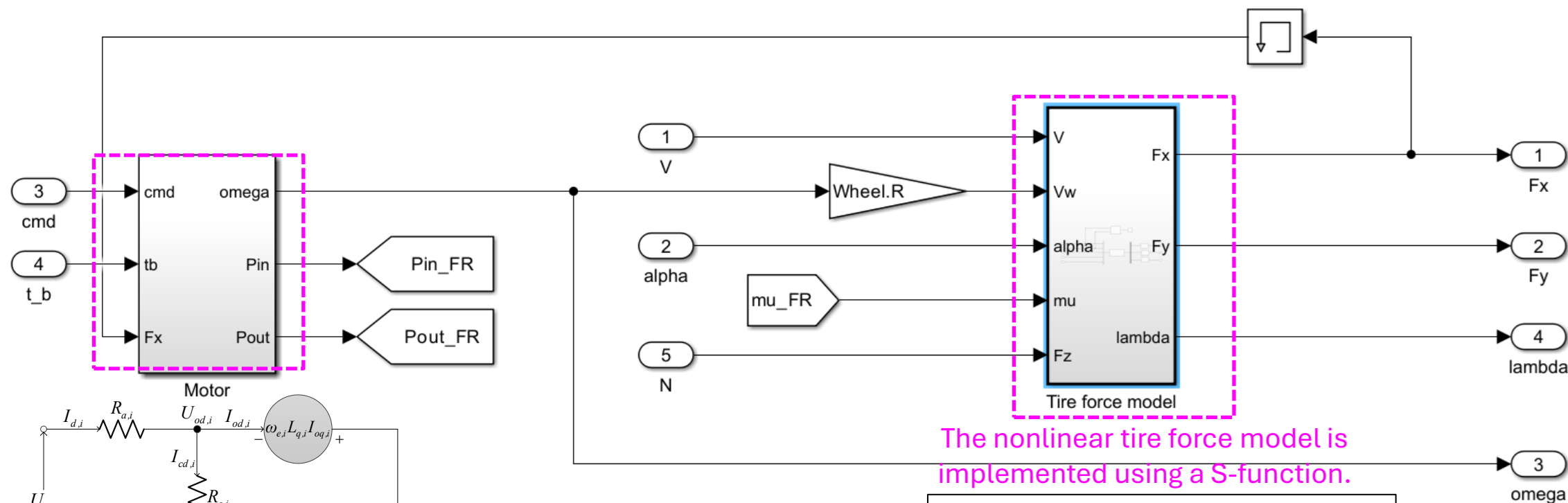
Simulator Ver. 002 (With Dynamics Wireless Power Transfer)



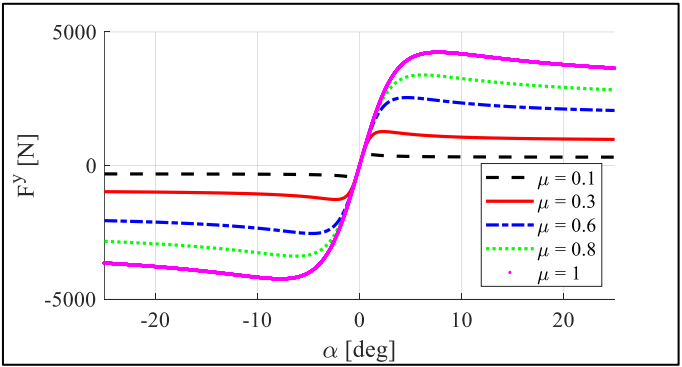
Simulation: Vehicle mode



Simulation: Wheel dynamics



The nonlinear tire force model is implemented using a S-function.



The d-q equivalent circuit of the PMSM

Examples of MESO strategies

Baseline strategy (BS):

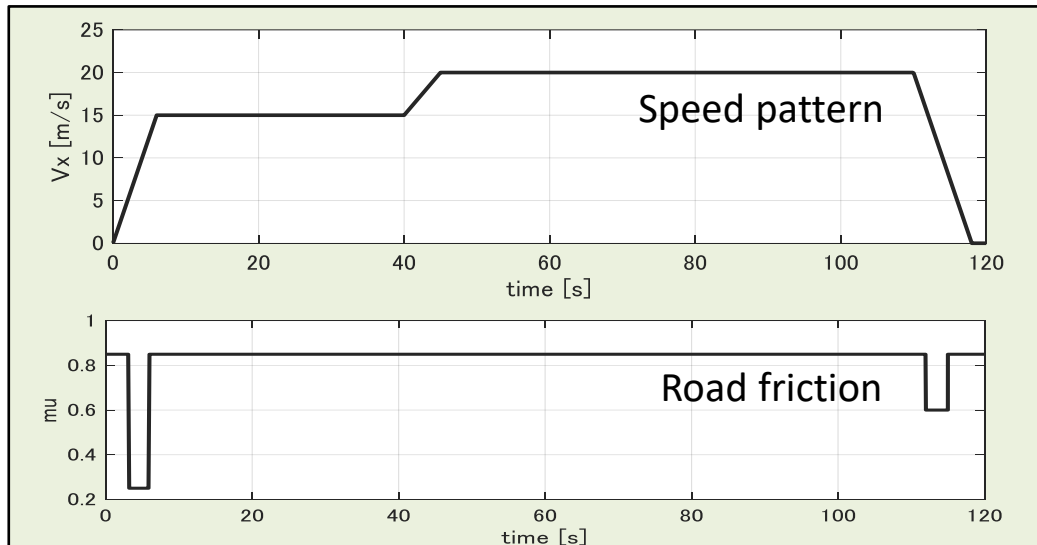
❖ Front EPS angle control:

$$\delta_f^* = K_{\delta y, p} e_y + K_{\delta \varphi, p} e_\varphi$$

where $e_y = y^* - y$, $e_\varphi = \varphi^* - \varphi$ with the reference values $y^* = 0$ and $\varphi^* = 0$. The control gains are $K_{\delta y, p}$ and $K_{\delta \varphi, p}$.

❖ IWM torque distribution: The torque distribution ratios are fixed as: $\{t_{1,2} = 0.27; t_{3,4} = 0.23\} \rightarrow$ no yaw moment is applied.

❖ Motor control: The d-axis current is $I_{d,i}^* = 0$ for all IWMs.



Improved strategy (IS):

❖ Front EPS angle control: The same as that of the BS.

❖ IWM torque distribution: The IWM torques are distributed according to a yaw moment control law:

$$\begin{cases} T_{1,3}^* = \frac{T_{all}^*}{4} - \frac{r}{2d} N^{z*}; & T_{2,4}^* = \frac{T_{all}^*}{4} + \frac{r}{2d} N^{z*} \\ N^{z*} = K_{Ny, p} e_y + K_{Ny, i} \int_0^t e_y dt + K_{Ny, d} \frac{de_y}{dt} + K_{N\varphi, p} e_\varphi \end{cases}$$

with the control gains $K_{Ny, p}$, $K_{Ny, i}$, $K_{Ny, d}$, and $K_{N\varphi, p}$.

❖ Motor control: Assuming that the motor salient coefficients are close to unity, the d-axis current that minimizes motor losses is given by the following formula (Morimoto *et al*, 1994):

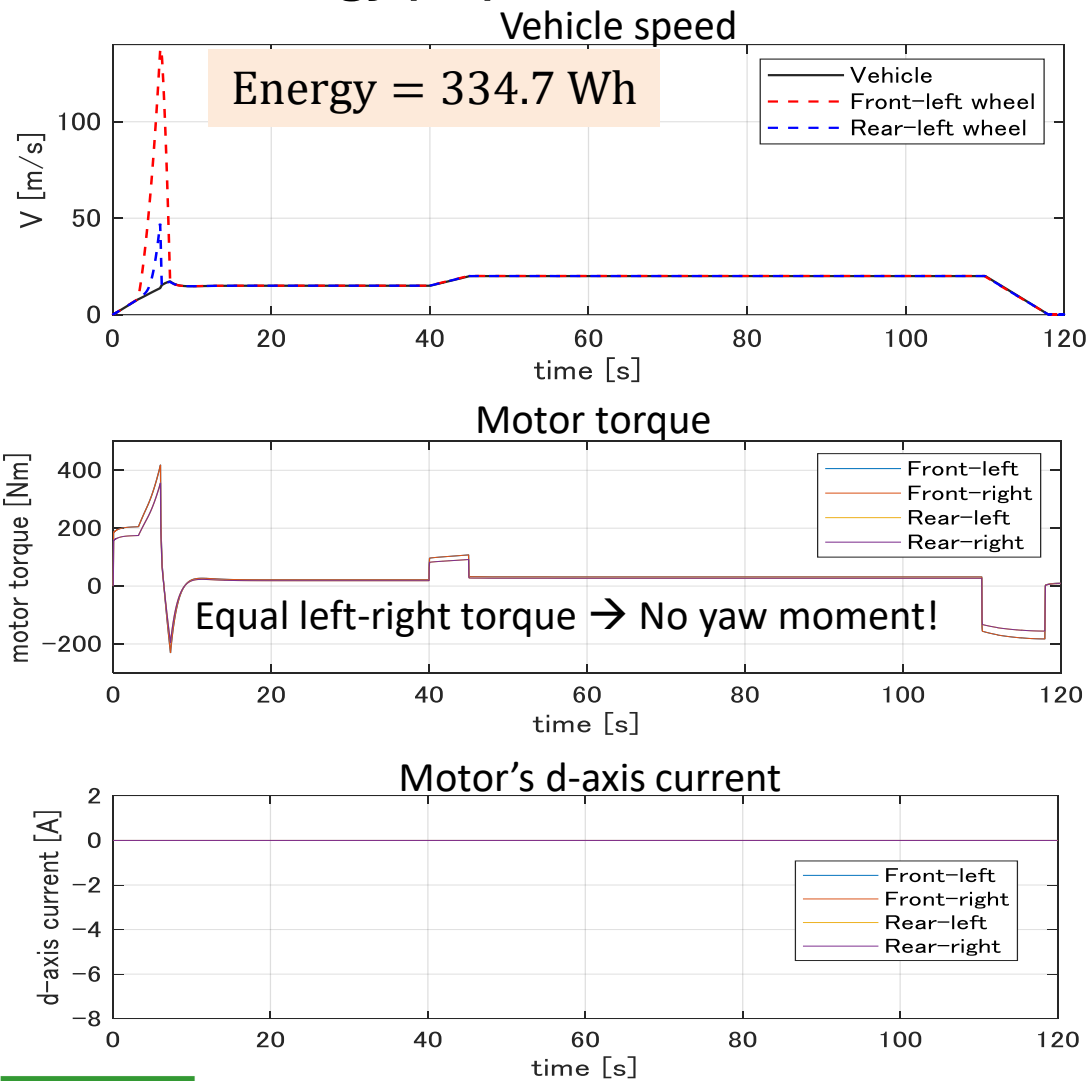
$$I_{od, i}^* = - \frac{\omega_{e, i}^2 L_{d, i} (R_{a, i} + R_{c, i}) \Psi_{a, i}}{R_{a, i} R_{c, i}^2 + \omega_{e, i}^2 L_{d, i}^2 (R_{a, i} + R_{c, i})}$$

S. Morimoto, Y. Tong, Y. Takeda and T. Hirasa, "Loss minimization control of permanent magnet synchronous motor drives," in *IEEE Transactions on Industrial Electronics*, vol. 41, no. 5, pp. 511-517, Oct. 1994, doi: 10.1109/41.315269.

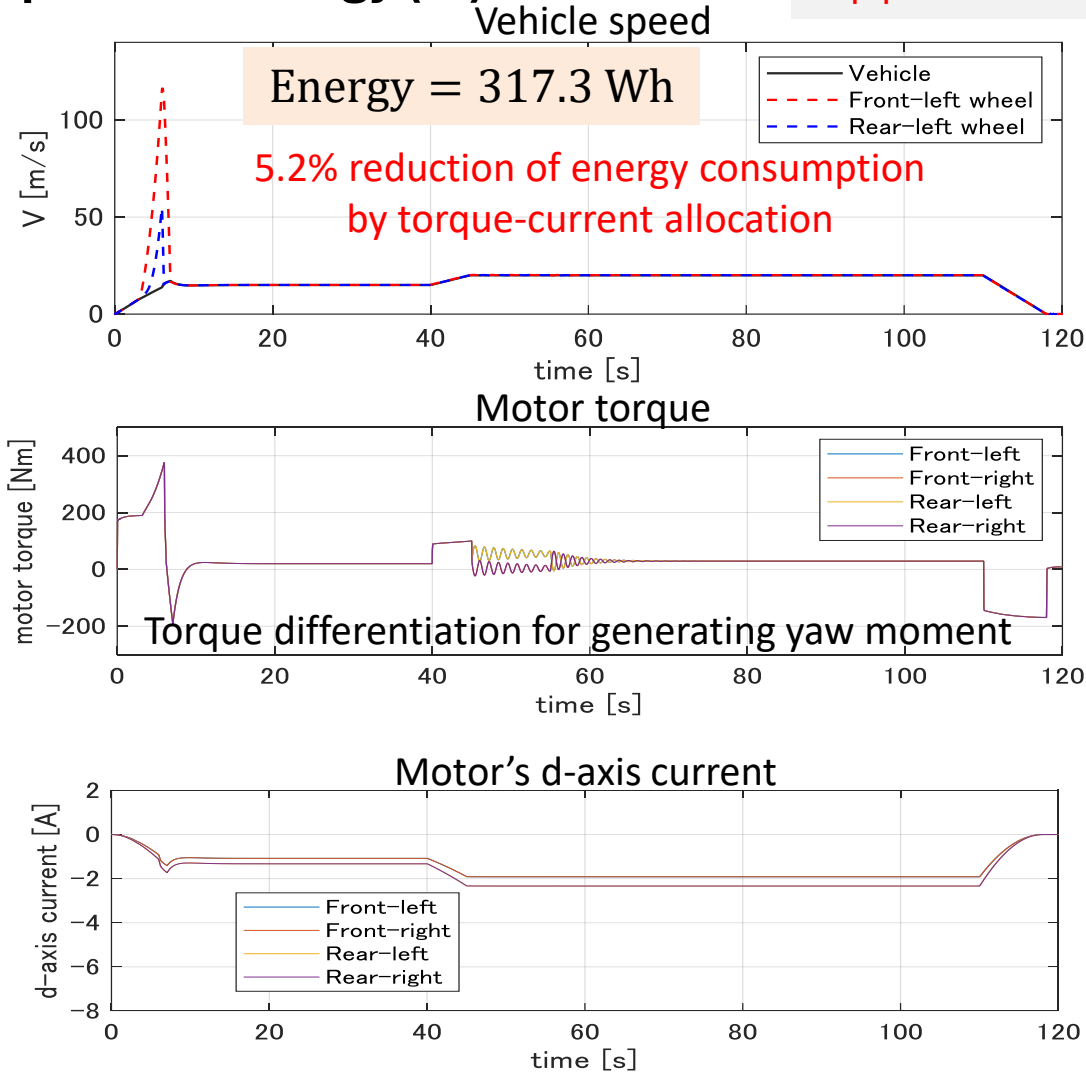
Simulation results

However, energy loss remains significant due to slip phenomenon.

Baseline strategy (BS)

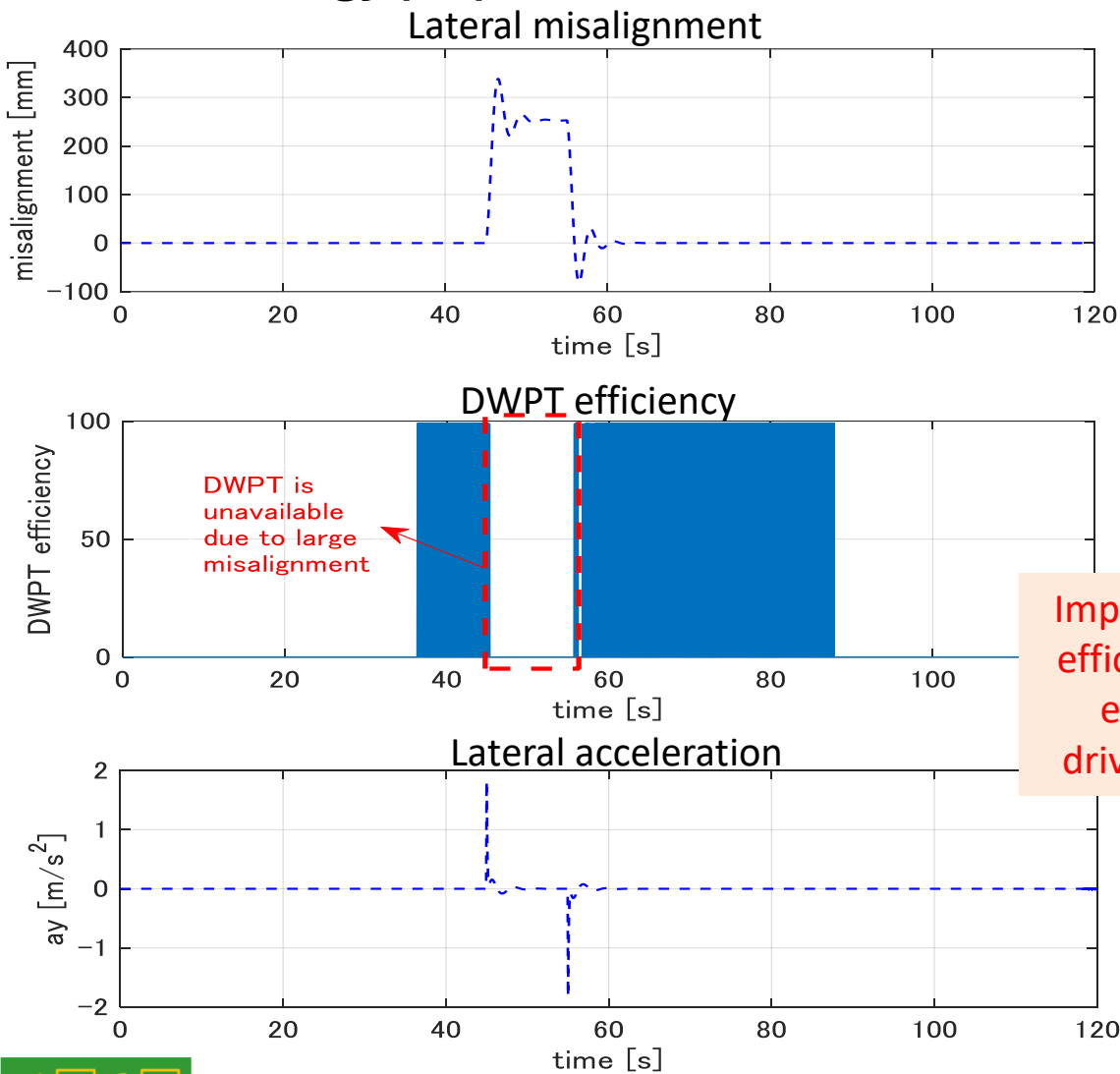


Improved strategy (IS)

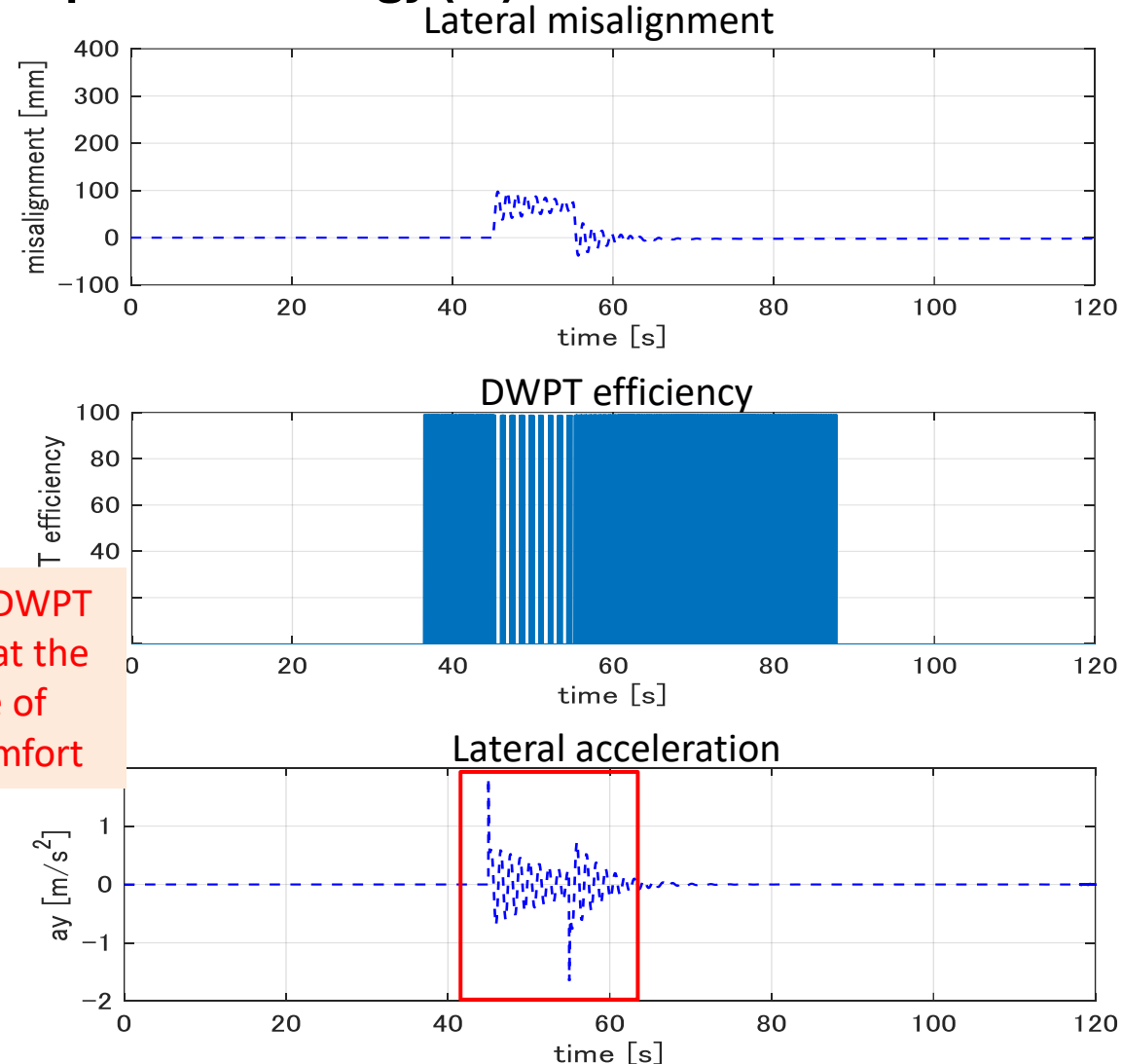


Simulation results

Baseline strategy (BS)



Improved strategy (IS)



Improved DWPT efficiency at the expense of driving comfort

Conclusions

Summary

- ❑ We have clarified the objectives, formulated the problem, and established the simulation for MVC 2027.
- ❑ Simulation results demonstrate the necessity of an integrated approach toward MESO:
 - ❖ Integration of longitudinal and lateral motion controls
 - ❖ Integration of driving force allocation and motor current allocation
 - ❖ Integration of WPT model and energy consumption model

Timeline

- ✓ 2025/10~2026/03: Problem formulation and simulator development
- ✓ 2026/04~2026/09: Preparation and final submission of MVC 2027 paper to VPPC 2026
- ✓ 2026/10: Presentation at VPPC 2026 and collection of feedback from research community
- ✓ 2026/10: Public of MVC 2027 on the website
- ✓ 2026/11~2027/01: Participant registration
- ✓ 2027/02: Submission of proposals by participants
- ✓ 2027/03: Evaluation and scoring of the submitted proposals

Thank you very much for your attention!

https://github.com/VTSociety/MVC_2027/



- Registration deadline: 31 January 2027
- Submission: 1 March 2027
- Results: 31 March 2027

The IEEE logo, featuring a diamond shape with a stylized 'I' and 'E' inside.**IEEE**

The VTS logo, with 'VTS' in a bold, italicized font and 'Connecting the Mobile World' in smaller text below it.**VTS**
Connecting the Mobile World

IEEE Vehicular Technology Society

The MESO logo, consisting of the letters M, E, S, and O in a stylized green font with yellow outlines.**MESO**A small red four-wheeled vehicle with a boxy, futuristic design, labeled '4IWM-EV' below it.**4IWM-EV**

A diagram showing a sequence of five 'DWPT Coil' units. The first unit is misaligned with the vehicle's path, indicated by a dashed line and the label 'Misalignment'. The subsequent units are aligned. A 'Strong lateral wind' arrow points towards the coils. The ground is labeled 'Low friction surface'.